

Monte Carlo Slot Engine & Volatility Simulator

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1 Executive Summary

This document outlines the mathematical framework and statistical validation for a 3-Reel Single Payline Casino Slot Engine. A 10,000,000 (10 Million) spin Monte Carlo simulation was executed to verify the empirical Return to Player (RTP), Hit Frequency, Volatility and Standard Deviation metrics against theoretical distributions.

2 Mathematical Framework & Formulas

1. Return To Player (RTP)

$$\text{RTP} = \left(\frac{\sum_{i=1}^N W_i}{N \times B} \right) \times 100$$

Where W_i = Win payout for spin i , N = Total Spins (10^7) and B = Base Bet (1.0).

2. Hit Frequency (HF)

$$\text{HF} = \left(\frac{\sum_{i=1}^N \mathbb{I}(W_i > 0)}{N} \right) \times 100$$

Where $\mathbb{I}$ is the indicator function evaluating to 1 if a spin results in a payout, and 0 otherwise.

3. Standard Deviation (σ – Volatility Index)

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (W_i - \mu)^2}$$

Where μ is the mean payout per spin.

4. 95% Confidence Interval Half-Width

$$\text{MoE}_{95\%} = 1.96 \times \left(\frac{\sigma}{\sqrt{N}} \right) \times 100$$

3 Reel Layout & Paytable Specification

- **Reel Stops:** 20 Stops per Reel.
- **Symbol Distribution:**
 - **7 (Jackpot):** 1 Stop per reel

- **B (Bar):** 3 Stops (Reels 1, 2), 2 Stops (Reel 3)
- **C (Cherry):** 4 Stops (Reels 1, 2), 3 Stops (Reel 3)
- **R (Respin):** 2 Stops (Reel 3)
- **_ (Blank):** 12 Stops per reel
- **Paytable Rules:**
 - $3 \times \text{'7'} = 200 \times \text{Bet}$
 - $3 \times \text{'B'} = 50 \times \text{Bet}$
 - $3 \times \text{'C'} = 20 \times \text{Bet}$
 - $1 \times \text{'C'} \text{ (Reel 1)} = 5 \times \text{Bet}$
 - **'R' Trigger (Reel 3) = Free Respin with $3 \times$ Payout Multiplier**

4 Empirical Simulation Results

Metric	Simulated Value
Total Simulated Spins	10,000,000
Base Game Bet	1.0 Unit per Spin
Simulated RTP	93.7627%
95% Confidence Margin of Error	$\pm$ 0.2462%
Hit Frequency	14.42%
Volatility (σ)	3.9729